

Nick Morse

Curriculum Vitae

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Stockholm, Sweden
Citizenship: USA

Researcher with expertise in large-eddy simulation (LES) and direct numerical simulation (DNS) of turbulent boundary layers, jets, and multiphase flows.

Education

- 2023 **PhD**, *Aerospace Engineering & Mechanics*, University of Minnesota
Thesis: "High-fidelity unstructured overset simulation of complex turbulent flows" [↗](#)
Adviser: Professor Krishnan Mahesh
- 2020 **MS**, *Aerospace Engineering & Mechanics*, University of Minnesota
- 2018 **BAEM**, *Aerospace Engineering & Mechanics*, University of Minnesota
Minors: Math, Astrophysics

Skills

<i>Coding</i>	Fortran, Python, Matlab, Bash, Git	<i>CFD</i>	Neko, OpenFOAM, ANSYS, Basilisk
<i>HPC</i>	MPI, Make, CMake, Hypr	<i>Meshing</i>	Pointwise, GridPro, Salome, Gmsh
<i>Visualization</i>	ParaView, Tecplot, VisIt, PyVista	<i>Engineering</i>	SolidWorks, Simulink
<i>Office</i>	L ^A T _E X, Microsoft Office	<i>Manufacturing</i>	Machining, waterjet, composites

Experience

- May 2025 – Present **Postdoctoral Researcher**, *KTH Royal Institute of Technology*, Stockholm, Sweden
- Analyzed structural characteristics of curved and rotating disk turbulent boundary layers.
 - Performed DNS with the open-source GPU-accelerated spectral element code *Neko* [↗](#).
- Jan – May 2025 **Group Leader: Multiscale CFD**, *Research Center Pharmaceutical Engineering*, Graz, Austria
- Aug 2023 – May 2025 **Senior Scientist**, *Research Center Pharmaceutical Engineering*, Graz, Austria
- Led the simulation strategy development for an EU Horizon 2020 project.
 - Coded a boundary element method from scratch to resolve sub-Kolmogorov-scale droplet breakup.
 - Discretization*: Unstructured Lagrange 6-point triangular cells, Gauss-Legendre quadrature, adaptive mesh refinement, feature-preserving Laplacian mesh smoothing.
 - Numerics*: Adaptive RK2 time stepping, matrix-free GMRES(k), fast multipole method.
 - I/O*: Binary VTK and Gmsh grid/solution files, automatic grid generation.
 - Implemented an ethanol-water mixture model in OpenFOAM to simulate impingement jet mixing.
- 2018 – 2023 **Graduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
Computational Fluids Lab (Professor Krishnan Mahesh)
- Performed large-scale (~10000 processor) simulations of complex turbulent flows using HPC.
 - Formulated streamline coordinates to study curvature in axisymmetric turbulent boundary layers.
 - Resolved experimental trip wires in LES to analyze model-scale boundary layer memory effects.
 - Identified mixing-enhancing secondary vortices from DNS and DMD of a jet in crossflow.
 - Extended an in-house unstructured finite-volume overset LES/DNS method in Fortran (MPI).
 - Implemented support for *hypr* GPU solvers with minimized LHS matrix reconstruction.
 - Added non-orthogonal Crank-Nicolson viscous flux correction and LES source terms.
 - Created grid cutting and hierarchy algorithms for overset assembly of grids for complex geometries.
- 2016 – 2017 **Undergraduate Research Assistant**, *University of Minnesota*, Minneapolis, MN, USA
Turbulent Shear Flow Lab (Professor Ellen Longmire)
- Designed tunable-buoyancy spheres to study particle transport in a turbulent boundary layer.
 - Characterized Stokes-number dependence of particle restitution using high-speed imaging.

2014–2019 **Chief Engineer & Aerodynamics Designer**, *University of Minnesota Formula SAE*

- Directed systems-level engineering design and led weekly meetings of a 70-member team.
- Programmed a Matlab graphical user interface to parameterize multi-element wing profiles.
- Automated large-scale ANSYS CFX simulations at the Minnesota Supercomputing Institute.

Service

2025 Co-organizer and session chair of a workshop at the *Banff International Research Station*: “Particulates across Scales: Mathematical Modeling, Computation, and Applications” [↗](#)

Publications

Journal articles

N. Morse. Insights into sub-Kolmogorov-scale droplet breakup via boundary element simulations. *International Journal of Multiphase Flow*, page 105460, 2025.

N. Morse and K. Mahesh. Tripping effects on model-scale studies of flow over the DARPA SUBOFF [↗](#). *Journal of Fluid Mechanics*, 975:A3, 2023.

N. Morse and K. Mahesh. Effect of tabs on the shear layer dynamics of a jet in cross-flow [↗](#). *Journal of Fluid Mechanics*, 958:A6, 2023.

N. Morse and K. Mahesh. Large-eddy simulation and streamline coordinate analysis of flow over an axisymmetric hull [↗](#). *Journal of Fluid Mechanics*, 926:A18, 2021.

Conference papers & abstracts

N. Morse. Insights into sub-Kolmogorov droplet breakup via a boundary element method [↗](#). *BIRS Workshop: Particulates across Scales: Mathematical Modeling, Computation, and Applications*, 2025.

N. Morse, J. Remmelgas, and J. Khinast. A simulation framework for nanodroplet breakup in top-down nanoparticle production [↗](#). *AIChE Annual Meeting*, 2024.

M. Fenelon, Y. Zhang, L. Cattafesta, **N. Morse**, K. Mahesh, L. Li, and Z. Pan. Optimized timing schemes for multi-pulse shake-the-box particle tracking velocimetry [↗](#). *AIAA SciTech Forum*, 2023.

N. Morse and K. Mahesh. The shear layer structure of a tabbed jet in crossflow [↗](#). *75th Annual Meeting of the APS DFD*, 2022.

M. Fenelon, L. Cattafesta, Y. Zhang, K. Mahesh, and **N. Morse**. Optimizing dt for MP-STB in particle tracking velocimetry [↗](#). *75th Annual Meeting of the APS DFD*, 2022.

N. Morse, T. Kroll, and K. Mahesh. Large-eddy simulation of submerged marine vehicles [↗](#). *Proceedings of the 34th Symposium on Naval Hydrodynamics, Washington, DC*, 2022.

N. Morse and K. Mahesh. Streamline coordinate analysis of the flow past an axisymmetric body computed by large-eddy simulation [↗](#). *74th Annual Meeting of the APS DFD*, 2021.

N. Morse and K. Mahesh. Large-eddy simulation of appended submerged vehicles using an unstructured overset grid method [↗](#). *73rd Annual Meeting of the APS DFD*, 2020.

T. Kroll, **N. Morse**, W. Horne, and K. Mahesh. Large eddy simulation of marine flows over complex geometries using a massively parallel unstructured overset method [↗](#). *Proceedings of the 33rd Symposium on Naval Hydrodynamics, Osaka, Japan*, 2020.

N. Morse, W. Horne, and K. Mahesh. Towards large-eddy simulation of maneuvering vehicles using an unstructured overset grid method [↗](#). *72nd Annual Meeting of the APS DFD*, 2019.

D. Barros, Y. H. Tee, **N. Morse**, B. Hiltbrand, and E. Longmire. Investigation of particle lift off in a turbulent boundary layer [↗](#). *70th Annual Meeting of the APS DFD*, 2017.

Interests

Mountain biking, road cycling, skiing, running, hiking, tennis, bouldering